

Juan David Muñoz-Bolaños

Rechenauerstraße 42, Innsbruck, Austria (Tirol)

+43 676 9115145

juan.munoz@i-med.ac.at

AMACO GmbH (Coesia Group)

Hiring Team

Millennium Park 10

6890 Lustenau, Austria

Subject: Application for Technical Project Engineer - Workflow and Vision Systems

Dear Hiring Team,

With hands-on experience deploying industrial vision pipelines and four years of integrated research engineering work in Austria, I am glad to apply for the Technical Project Engineer, Workflow and Vision Systems position at AMACO in Lustenau.

As a Junior Machine Vision Project Engineer at INTECOL S.A.S., I helped the implementation of a multi-station inspection system on a high-speed glass bottle line, designing bright-field, dark-field, and ring-light illumination schemes, configuring Halcon pipelines for label verification, liquid level, cap sealing, and print quality, and driving the project through to full customer sign-off. That experience sharpened my ability to balance optical physics, system robustness, and on-time delivery at the customer/engineering interface.

After the experience of the real world and industry, I wanted to pursue a PhD. My research further developed these skills through end-to-end C++/Python system integration, scientific hardware interfacing, and real-time pipeline development with product results published in peer-reviewed journals. I adapt quickly to new toolchains, communicate technical complexity clearly to non-specialist stakeholders, and am fully integrated into the Austrian professional environment (B2 German). Available from May 2026 since my contract ends; my current Resident Permit can be converted seamlessly to a work permit.

Sincerely,

Juan David Muñoz-Bolaños

Vision and Automation Project Engineer

Medical University of Innsbruck

Juan David Muñoz-Bolaños

Vision and Automation Project Engineer

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PROFESSIONAL SUMMARY

Machine vision and automation engineer with hands-on industrial experience commissioning Halcon-based inspection systems and four years of end-to-end optical system development in Austria. Proven track record delivering projects from hardware specification and illumination design through software integration, validation, and customer sign-off. Proficient in C++, Python, and LabVIEW; fast learner on new vision toolchains. B2 German; available May 2026.

TECHNICAL COMPETENCIES

Machine Vision and Inspection

Hands-on deployment of Halcon-based multi-station inspection systems on production lines through full customer commissioning. Experienced in camera selection, custom illumination design, and pipeline optimisation for high-speed environments.

Project Engineering

Assisted in managing the full lifecycle of a machine vision project: hardware specification, illumination design, software integration, validation, and customer sign-off. PhD project management across hardware, software, and international vendors (Thorlabs, Hamamatsu, Edmund Optics).

Software and Automation

Proficient in C++, Python, and LabVIEW for real-time hardware control, data acquisition, and automated calibration. Fast learner on new vision platforms (Halcon, OpenCV, Cognex). Applied JAX for GPU-accelerated phase retrieval in published research.

Communication and Training

Fluent English; B2 German (certificate). Skilled at translating complex optical and software systems into clear documentation and operator-facing explanations. Selected for ASML and ZEISS Summer Schools related to competitive industry recognition.

PROFESSIONAL EXPERIENCE

Medical University of Innsbruck

Innsbruck, Austria

PhD Researcher - Optical Systems and Automation

Jan 2022 – Present

- **Adaptive Optics Microscope (Biomedical Optics Express, 2023):** Led the complete hardware-to-software build of a multiphoton nonlinear microscope, including laser source integration, galvo-scanner control, PMT signal acquisition, and adaptive optics correction, achieving diffraction-limited resolution.
- **Wavefront Sensor and Phase Retrieval System (ACS Photonics, 2025):** Designed and built from scratch an automated wavefront sensing system for adaptive optics microscopy. Wrote the full C++/Python/LabVIEW control stack, interfaced scientific cameras and deformable mirrors, and implemented iterative phase retrieval algorithms. System is in daily use and central to two peer-reviewed publications.
- **ZEMAX Tolerance Analysis and Optical Simulation:** Characterised aberration sensitivity of custom microscope objectives through systematic tolerance analysis in ZEMAX alongside wave-field simulations in Python (Zernike decomposition), directly applicable to vision system optical design.
- **Lab Management and Supplier Coordination:** Maintained health, safety, and environment compliance, calibration, and procurement logs for a high-power laser lab over 4 years. Managed component sourcing from Thorlabs, Hamamatsu, and Edmund Optics with incoming inspection.

Leibniz Institute of Photonic Technology (IPHT)

Jena, Germany

R and D Researcher - Photonics and Data Analysis

Jun 2020 – Dec 2021

- **1064 nm Fiber Probe Raman Spectrometer (Applied Sciences, 2024):** Designed and built a dispersive fiber probe Raman spectrometer for non-invasive human tissue characterisation, covering optical layout, detector calibration, and fiber coupling, from prototype to a published clinical instrument.
- **Supervised ML on Hyperspectral Tissue Images:** Applied scikit-learn classifiers to 2D hyperspectral Raman images of human tissue, achieving high-accuracy separation of cancerous from healthy cells. Accelerated the experimental characterisation loop significantly.

INTECOL S.A.S.

Medellín, Colombia

Junior Machine Vision Project Engineer

Mar 2018 – Dec 2018

- **Multi-Station Halcon Inspection System (Customer Sign-Off):** Assisted in the engineering and commissioning of an inspection system on a high-speed glass bottle line covering label verification, liquid level control, cap sealing, and print quality. Designed custom bright-field, dark-field, and ring-light illumination for each station. Client signed off on full production deployment.
- **Hardware Specification and Integration:** Selected industrial cameras, lenses, lighting hardware; built C++ Halcon pipelines tuned to the throughput and reliability requirements of the live production environment.

EDUCATION

Medical University of Innsbruck

Austria

Ph.D. in Adaptive Optics & Microscopy

Jan 2022 – Present

Friedrich Schiller University Jena

Germany

M.Sc. in Photonics

2019 – 2021

University of Cauca

Colombia

B.Sc. in Physics Engineering

2012 – 2018

PUBLICATIONS & AWARDS

- **Awards and Selections:** Selected for ASML and ZEISS Summer Schools; awarded COLFUTURO Grant 2026 for best Colombian professionals.
- Muñoz-Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* 12(1), 176-184 (2025).
- Muñoz-Bolaños, J. D. et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer. *Applied Sciences*, 14(11), 4726 (2024).
- Sohmen, M., Muñoz-Bolaños, J. D., et al. Optofluidic adaptive optics in multi-photon microscopy. *Biomedical Optics Express* 14(4), 1562-1578 (2023).

LANGUAGES

English (Fluent / Professional),
German (B2 - Working Proficiency),
Spanish (Native)

REFERENCES

Prof. Monika Ritsch-Marte

Medical University of Innsbruck

Head of Institute for Biomedical Physics

monika.ritsch-marte@i-med.ac.at

Dr. Christoph Krafft

Leibniz Institute of Photonic Technology

Group leader: Spectroscopy Imaging

christoph.krafft@leibniz-ipht.de

AGREEMENT FOR GUESTS

to carry out a practical training
with

Mr. **Juan David Munoz Bolanos** Date of birth: **1994-06-04**
temporarily living at **Emil-Wölk-Str. 7, 07747 Jena** (guest),

Student of the Friedrich-Schiller-University Jena, Abbe School of Photonics, gets in execution of his practical course in the research department Spectroscopy and Imaging, working group Raman and Infrared Histopathology (0104) from 2020-06-02 up to 2020-12-31 admission to the necessary rooms within Leibniz-IPHT.

The use of further devices, media and other infrastructure takes place in agreement with the Heads of the respective departments.

Mr. Juan David Munoz Bolanos has to observe the security regulations for working and other regulations valid in Institute as well as instructions of the laboratory and clean room responsible persons of the Leibniz-IPHT.

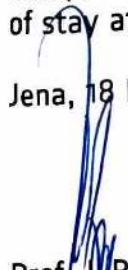
Mr. Juan David Munoz Bolanos has to treat all affairs and procedures strictly confidential. Copies, duplicates and others of business and technical information of each kind are forbidden, unless these are sanctioned by the responsible Department Heads.

Publications, reports and lectures about topics which belong together with the fields of work of Leibniz-IPHT require the previous written agreement of the Executive Board.

The liability of Mr. Juan David Munoz Bolanos is limited to cases of wilful action and gross negligence.

Mr. Juan David Munoz Bolaos is insured against accidents on the Friedrich-Schiller-University Jena, Abbe School of Photonics in the context of his/her practical course during his/her length of stay at Leibniz-IPHT.

Jena, 18 May 2020


Prof. Dr. Popp
Chairman
of Leibniz-IPHT


F. Sonderrmann
Executive Member
of Leibniz-IPHT


Juan David Munoz Bolanos
Guest

INTECOL

OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

TO WHOM IT MAY CONCERN

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos (\$1,500,000) plus three hundred thousand pesos (\$300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

INTECOL

NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO

Administrative and Financial Director

Integración Tecnológica Industrial Intecol S.A.S.
www.intecol.com.co
Medellin Calle 42 #63c-82
PBX. (574) 316 3322 Cell: (57) 301 430 2532



URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die
Physikalisch-Astronomische Fakultät
mit dieser Urkunde

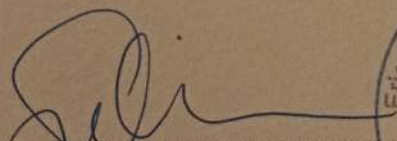
Herrn Juan David Munoz Bolanos
geb. 04.06.1994 in La Cruz (Kolumbien)
den Hochschulgrad

MASTER OF SCIENCE (M.Sc.)

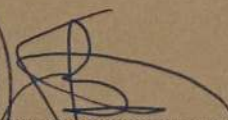
Er hat die Masterprüfung (120 ECTS)
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021


Prof. Dr. Christian Spielmann
Dekan




Prof. Dr. Silvana Botti
Vorsitzende des
Prüfungsausschusses

OFFICIAL TRANSLATION OF DOCUMENTS WRITTEN IN SPANISH, MADE 20
DECEMBER 2018 BY JUAN CARLOS OJEDA-GARCES, OFFICIAL TRANSLATOR
BY RESOLUTION No. 1171 ISSUED BY THE MINISTRY OF JUSTICE, REPUBLIC
OF COLOMBIA.



UNIVERSIDAD DEL CAUCA
ON BEHALF OF THE
REPUBLIC OF COLOMBIA
AN BY AUTHORIZATION OF THE MINISTRY OF NATIONAL EDUCATION
CONSIDERING THAT

JUAN DAVID MUÑOZ BOLAÑOS
CC 1.061.772.278 OF POPAYAN

HAS COMPLIED WITH ALL THE STATUTORY AND LEGAL REQUIREMENTS,
GRANTS THE TITLE OF

PHYSICAL ENGINEER

WITH ALL THE RIGHTS, PRIVILEGES AND DIGNITIES THAT AUTHORIZE HIM
FOR THE PROFESSIONAL EXERCISE

POPAYAN, 16 MARCH 2018

REGISTERED IN THE DIPLOMA BOOK No. 082 FOLIO 437 DIPLOMA No. 396
RESOLUTION No. R-266-07 MINUTE No. 14-07

(SIGNED)

THE PRINCIPAL OF THE UNIVERSITY
THE DEAN OF THE FACULTY
THE SECRETARY GENERAL OF THE UNIVERSITY

Juan Carlos Ojeda - Garcés
TRADUCTOR INTERPRETE OFICIAL
RESOLUCIÓN No 1171
MINISTERIO DE JUSTICIA 1997

Juan Munoz Bolanos

has participated in the

**ZEISS European
Summer School**

Lithography Optics &
Semiconductor Technologies

26 - 27 June 2025

ZEISS Semiconductor Manufacturing Technology, Oberkochen, Germany

Thomas Marschner

Dr. Thomas Marschner
Program Manager Funding Projects

Thomas Stämmler

Dr. Thomas Stämmler
CTO & Head of Product Strategy



Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ö s d

österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291

Zertifikat

Juan David MUÑOZ BALAÑOS

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Mündliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 11.08.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ösd

österreich schweiz deutschland



ID-Nummer: ZB2Ö25m40434

Passaporto № / Passport No

MUNOZ BOLANOS

COLOMBIANA

Sexo / Sex
Lugar de nacimiento / Place of birth

Fecha de expedición / Date of issue

Fecha de vencimiento / Date of expiry

G. ANTIOQUIA

Forma dei tubuli / *Peperideris* e *Agropyron*

Dear Family

[illegible]

